VIT BHOPAL

vityarthi
YOUR LEARNING DESTINATION

PROJECT REPORT

TITLE : Employee Login System with CAPTCHA Verification and Voice Feedback

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Course : Fundamental in AI & ML

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1. Introduction :

The increasing reliance on online platforms has made security verification essential. CAPTCHA (Completely Automated Public Turing test to tell Computers and Humans Apart) systems are widely used to distinguish human users from automated bots. This project focuses on designing and implementing a standalone, accessible CAPTCHA verification tool with enhanced usability and inclusivity features

2. Objectives :

- *Develop a GUI-based CAPTCHA system using Python (Tkinter).*
- *integrate credential verification for secure access.*
- *Provide accessibility features such as speech feedback for visually impaired users.*
- *Ensure modular, reusable, and human-readable code.*
- *Create clear documentation for user guidanc*

3. Features :

- *Random CAPTCHA generation (alphanumeric sequences).*
- *Input-driven verification with credential checks.*
- *Audio feedback for inclusivity.*
- *Threading integration to prevent GUI blocking.*
- *Modular codebase for scalability and reus*

4. Implementation :

- **Language**: Python 3.x
- **Libraries Used**: *Tkinter (GUI), threading (non-blocking execution), pyttsx3 (speech synthesis).*
- **Coding Style**: *Human-readable variable names, inline comments, and modular functions.*
- **Accessibility**: *Speech feedback for CAPTCHA characters and verification result*

5. Testing & Results :

- **Functional Testing:** Verified CAPTCHA generation, input validation, and credential checks.

- **Accessibility Testing:** Confirmed speech feedback works without GUI blocking.

- **Usability Testing:** Ensured intuitive interface and clear instructions for users.

Figure 5: Verification Success Message (screenshot placeholder)

Figure 6: Verification Failure Message (screenshot placeholder)

Figure 5 :

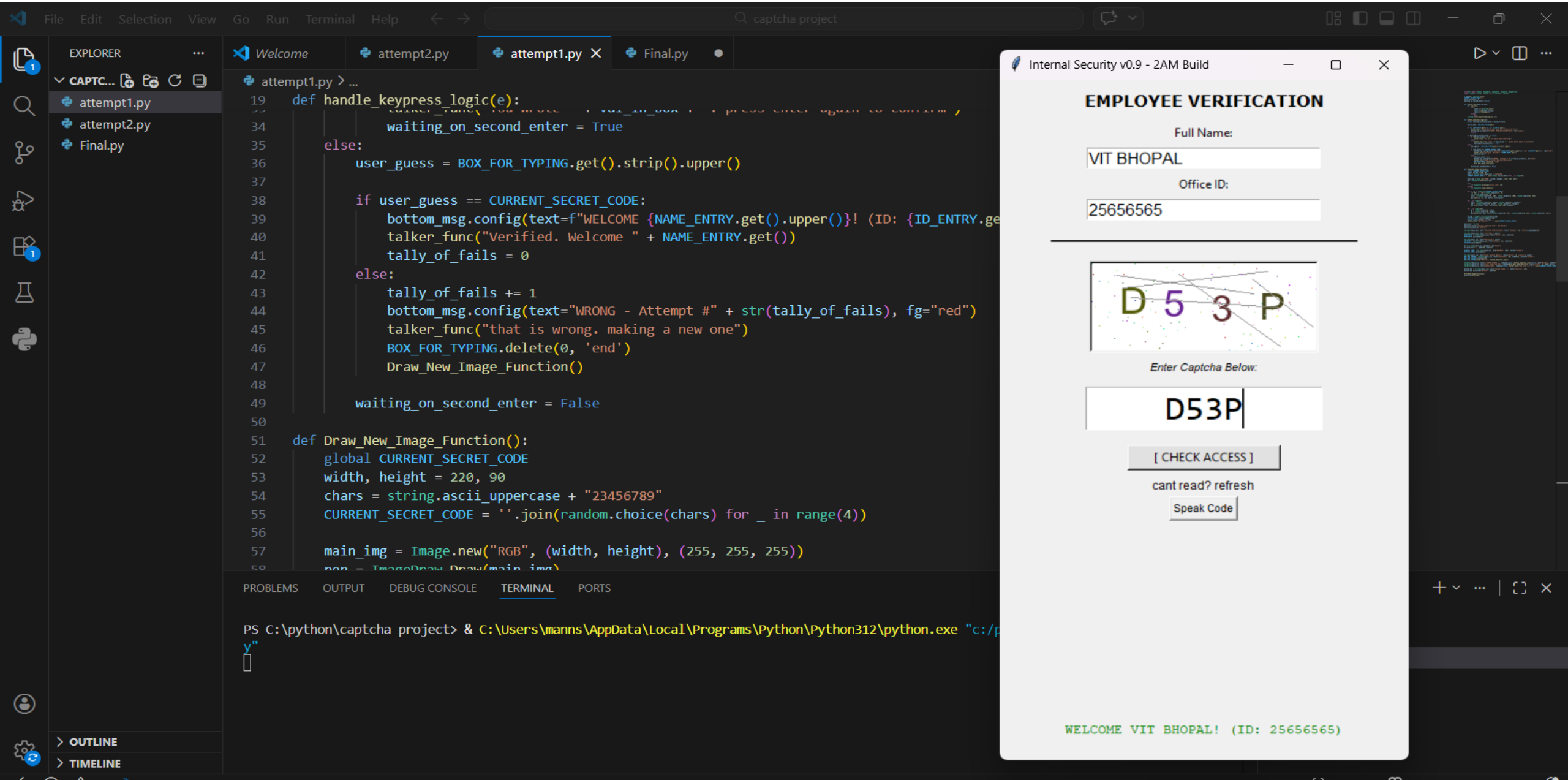
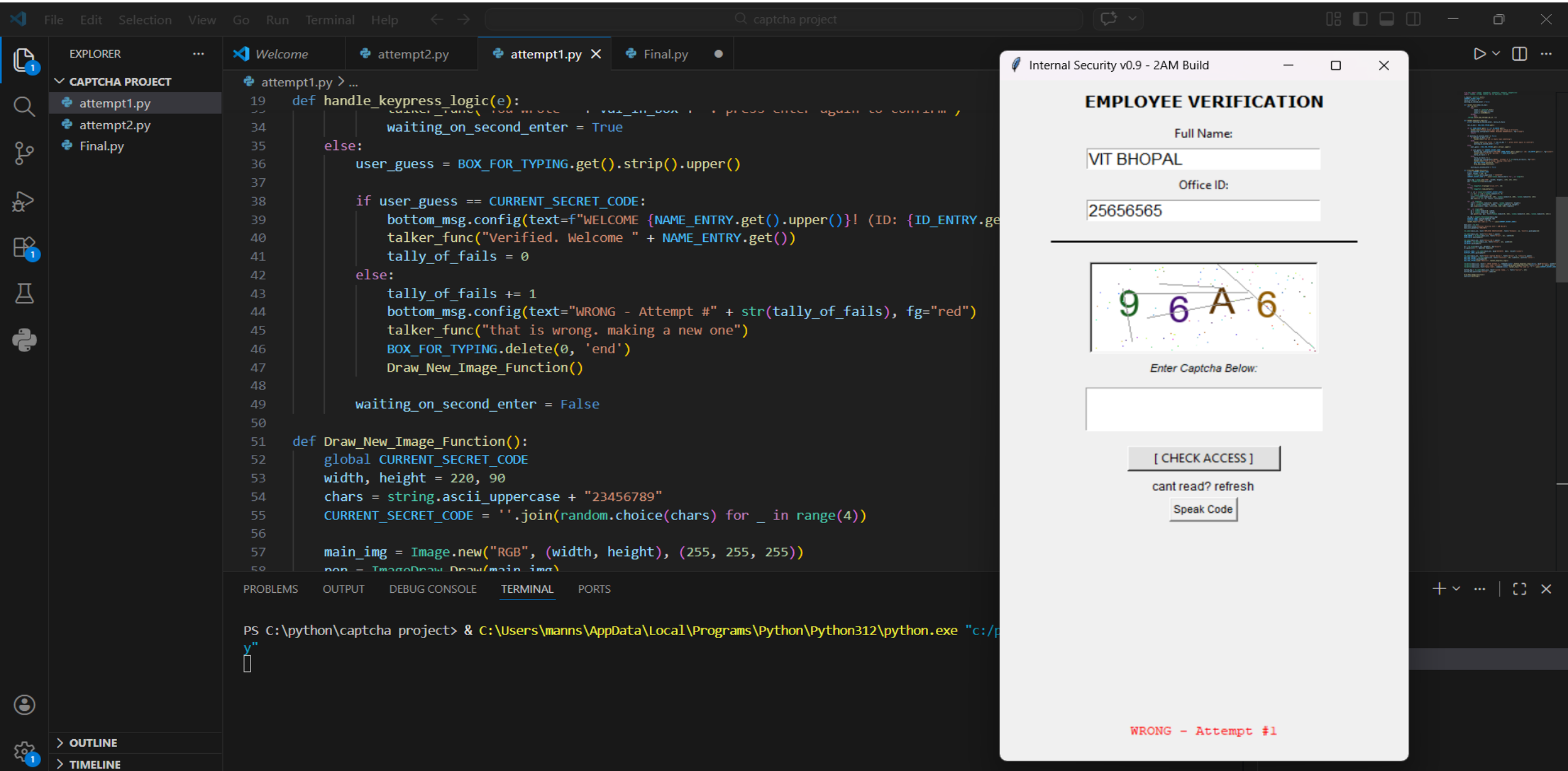

File Edit Selection View Go Run Terminal Help
captcha project
EXPLORER
CAPTC...
attempt1.py
attempt2.py
Final.py
Welcome
attempt2.py
attempt1.py
Final.py
attempt1.py > ...
19 def handle_keypress_logic(e):
34 waiting_on_second_enter = True
35 else:
36 user_guess = BOX_FOR_TYPING.get().strip().upper()
37
38 if user_guess == CURRENT_SECRET_CODE:
39 bottom_msg.config(text=f"WELCOME {NAME_ENTRY.get().upper()}! (ID: {ID_ENTRY.ge
40 talker_func("Verified. Welcome " + NAME_ENTRY.get())
41 tally_of_fails = 0
42 else:
43 tally_of_fails += 1
44 bottom_msg.config(text="WRONG - Attempt #" + str(tally_of_fails), fg="red")
45 talker_func("that is wrong. making a new one")
46 BOX_FOR_TYPING.delete(0, 'end')
47 Draw_New_Image_Function()
48
49 waiting_on_second_enter = False
50
51 def Draw_New_Image_Function():
52 global CURRENT_SECRET_CODE
53 width, height = 220, 90
54 chars = string.ascii_uppercase + "23456789"
55 CURRENT_SECRET_CODE = ''.join(random.choice(chars) for _ in range(4))
56
57 main_img = Image.new("RGB", (width, height), (255, 255, 255))
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\python\captcha project> & C:\Users\manns\AppData\Local\Programs\Python\Python312\python.exe "c:/p
y"
OUTLINE
TIMELINE
Internal Security v0.9 - 2AM Build
EMPLOYEE VERIFICATION
Full Name:
VIT BHOPAL
Office ID:
25656565
D 5 3 P
Enter Captcha Below:
D53P
[CHECK ACCESS]
cant read? refresh
Speak Code
WELCOME VIT BHOPAL! (ID: 25656565)

Figure 6: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EMPLOYEE VERIFICATION
Full Name:
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9 6 A 6
Enter Captcha Below:
[CHECK ACCESS]
cant read? refresh
Speak Code
WRONG - Attempt #1

6. Challenges :

- *Secure login systems.*
- *Educational demonstrations of CAPTCHA logic.*
- *Accessibility-focused software projects.*
- *Modular integration into larger authentication systems.*

7. Applications:

- *Secure login systems.*
- *Educational demonstrations of CAPTCHA logic.*
- *Accessibility-focused software projects.*
- *Modular integration into larger authentication systems*

8. Conclusion :

The project successfully delivers a flexible, accessible CAPTCHA verification tool. It balances security with inclusivity, ensuring usability for diverse users. The modular design and clear documentation make it adaptable for future enhancements.

9. Future Scope :

- Integration with databases for user management.
- Support for image-based CAPTCHAs.
- Cloud deployment for real-world applications.